

STAT4911_EDA_Account_B

Srinika Kalavala

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```
dataset <- read.csv("/Users/srinika/Desktop/OSU/Junior Year/Spring/STAT 4911/Battery  
Project/battery_dataset_20260327.csv")  
library(ggplot2)  
library(dplyr)
```

```

account_b_data <- dataset[dataset$AccountNumber == "account_b", ]
account_b_data$Month <- as.character(account_b_data$Month)

account_b_data$Datetime_parsed <- as.POSIXct(account_b_data$Datetime,
                                              format = "%m/%d/%y %H:%M")
account_b_data$Hour <- as.integer(format(account_b_data$Datetime_parsed, "%H"))

account_b_data$SettlementPointPrice <- account_b_data$SettlementPointPrice / 1000

account_b_data$Datetime_parsed <- as.POSIXct(account_b_data$Datetime, format = "%m/%
d/%y %H:%M")
account_b_data$Hour <- as.integer(format(account_b_data$Datetime_parsed, "%H"))

account_b_data$Season <- "Fall"
account_b_data$Season[account_b_data$Month %in% c("December", "January", "February")]
<- "Winter"
account_b_data$Season[account_b_data$Month %in% c("March", "April", "May")]
<- "Spring"
account_b_data$Season[account_b_data$Month %in% c("June", "July", "August")]
<- "Summer"

account_b_data$TimeOfDay <- "Night"
account_b_data$TimeOfDay[account_b_data$Hour >= 5 & account_b_data$Hour < 12] <- "Mo
rning"
account_b_data$TimeOfDay[account_b_data$Hour >= 12 & account_b_data$Hour < 17] <- "Af
ternoon"
account_b_data$TimeOfDay[account_b_data$Hour >= 17 & account_b_data$Hour < 20] <- "Ev
ening"

results <- aggregate(Quantity ~ Month + Season + TimeOfDay,
                     data = account_b_data,
                     FUN = sum)

month_levels <- c("January", "February", "March", "April", "May", "June",
                  "July", "August", "September", "October", "November", "December")
results$Month <- factor(results$Month, levels = month_levels)
results <- results[order(results$Month), ]

print(results)

```

##	Month	Season	TimeOfDay	Quantity
## 12	January	Winter	Afternoon	68626.71
## 24	January	Winter	Evening	35926.92
## 36	January	Winter	Morning	125002.26
## 48	January	Winter	Night	96864.34
## 11	February	Winter	Afternoon	50857.28
## 23	February	Winter	Evening	25603.07
## 35	February	Winter	Morning	95466.12
## 47	February	Winter	Night	67053.30
## 5	March	Spring	Afternoon	43268.31
## 17	March	Spring	Evening	19899.95
## 29	March	Spring	Morning	80909.61
## 41	March	Spring	Night	52981.72
## 4	April	Spring	Afternoon	37856.23
## 16	April	Spring	Evening	16734.36
## 28	April	Spring	Morning	61134.41
## 40	April	Spring	Night	39049.31
## 6	May	Spring	Afternoon	41265.24
## 18	May	Spring	Evening	17935.10
## 30	May	Spring	Morning	53865.26
## 42	May	Spring	Night	37232.81
## 9	June	Summer	Afternoon	57500.94
## 21	June	Summer	Evening	24601.81
## 33	June	Summer	Morning	72746.84
## 45	June	Summer	Night	46280.65
## 8	July	Summer	Afternoon	62709.48
## 20	July	Summer	Evening	27769.74
## 32	July	Summer	Morning	83460.54
## 44	July	Summer	Night	49428.56
## 7	August	Summer	Afternoon	68707.99
## 19	August	Summer	Evening	27784.99
## 31	August	Summer	Morning	82159.68
## 43	August	Summer	Night	47818.79
## 3	September	Fall	Afternoon	57909.97
## 15	September	Fall	Evening	23001.77
## 27	September	Fall	Morning	67878.13
## 39	September	Fall	Night	43800.58
## 2	October	Fall	Afternoon	48158.62
## 14	October	Fall	Evening	19776.09
## 26	October	Fall	Morning	64795.62
## 38	October	Fall	Night	45658.54
## 1	November	Fall	Afternoon	52266.95
## 13	November	Fall	Evening	24816.33
## 25	November	Fall	Morning	85101.43
## 37	November	Fall	Night	60853.52
## 10	December	Winter	Afternoon	63738.47
## 22	December	Winter	Evening	32523.98
## 34	December	Winter	Morning	109972.61
## 46	December	Winter	Night	80226.16

```

results_b <- aggregate(Quantity ~ Month + Season + TimeOfDay,
                        data = account_b_data, FUN = sum)

month_levels <- c("January", "February", "March", "April", "May", "June",
                  "July", "August", "September", "October", "November", "December")

results_b$Month <- factor(results_b$Month, levels = month_levels)
results_b$TimeOfDay <- factor(results_b$TimeOfDay,
                              levels = c("Morning", "Afternoon", "Evening", "Night"))
results_b <- results_b[order(results_b$Month), ]

```

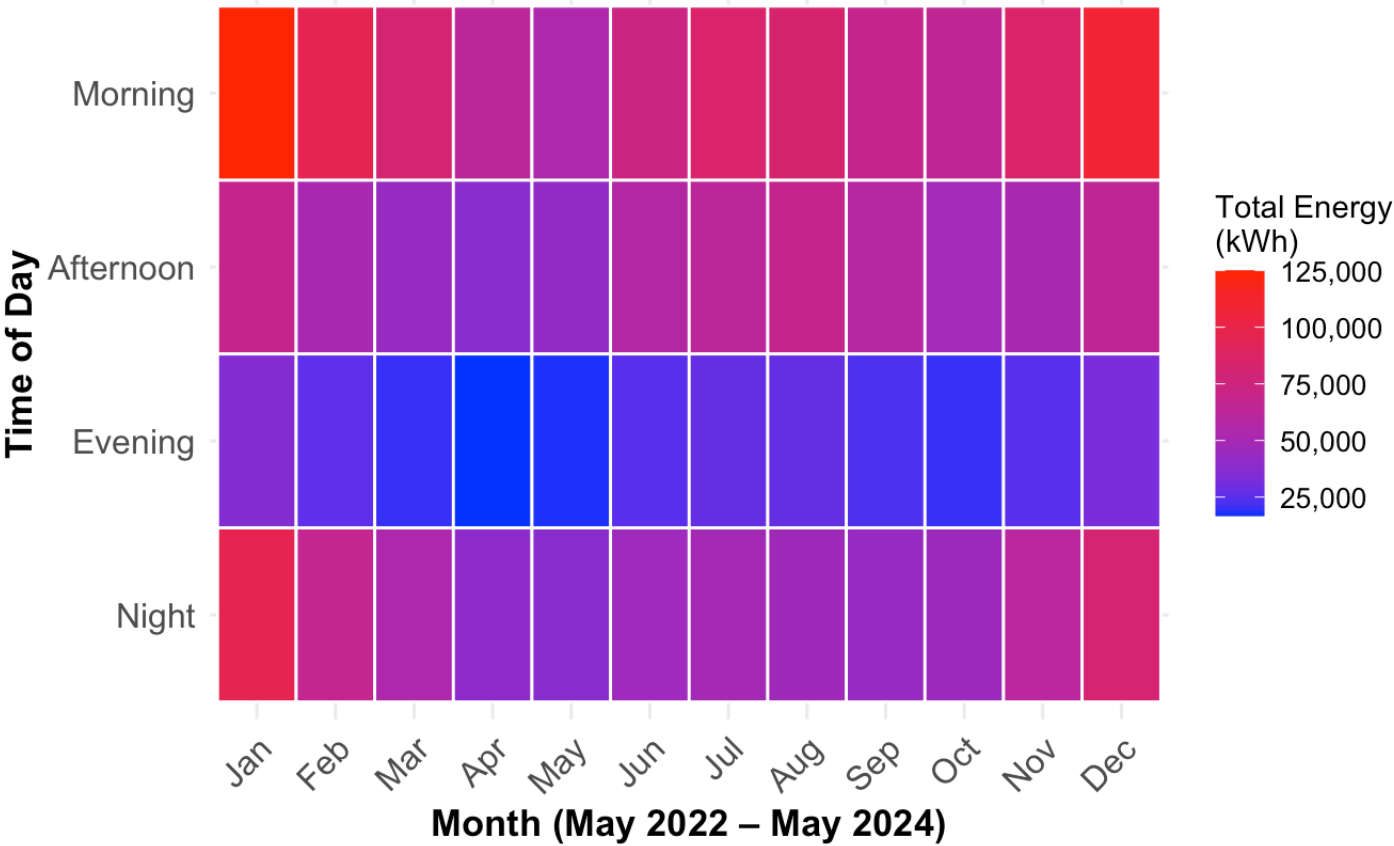
```

ggplot(results_b, aes(x = Month, y = TimeOfDay, fill = Quantity)) +
  geom_tile(color = "white", linewidth = 0.5) +
  scale_fill_gradient(
    low = "blue", high = "red",
    labels = scales::comma,
    name = "Total Energy\n(kWh)"
  ) +
  scale_x_discrete(labels = month.abb) +
  scale_y_discrete(limits = c("Night", "Evening", "Afternoon", "Morning")) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(size = 12, angle = 45, vjust = 1, hjust = 1),
    axis.text.y = element_text(size = 12),
    axis.title.x = element_text(size = 13, face = "bold"),
    axis.title.y = element_text(size = 13, face = "bold"),
    plot.title = element_text(size = 14, face = "bold"),
    plot.subtitle = element_text(size = 11, color = "gray40"),
    plot.caption = element_text(size = 10, color = "gray50"),
    legend.text = element_text(size = 10),
    legend.title = element_text(size = 11)
  ) +
  labs(
    title = "Account B: Total Hourly Energy Consumption Heatmap (May 2022 – May 2024)",
    subtitle = "Red = high total consumption; Blue = low total consumption",
    x = "Month (May 2022 – May 2024)",
    y = "Time of Day",
    caption = "Aggregated sum of hourly kWh readings per month and time-of-day block"
  )

```

Account B: Total Hourly Energy Consumption Heatmap (May 2022 - May 2024)

Red = high total consumption; Blue = low total consumption



Aggregated sum of hourly kWh readings per month and time-of-day block

```

account_b_data <- dataset[dataset$AccountNumber == "account_b", ]
account_b_data$Month <- factor(as.character(account_b_data$Month),
                              levels = c("January", "February", "March", "April",
                                           "May", "June", "July", "August",
                                           "September", "October", "November", "December"))

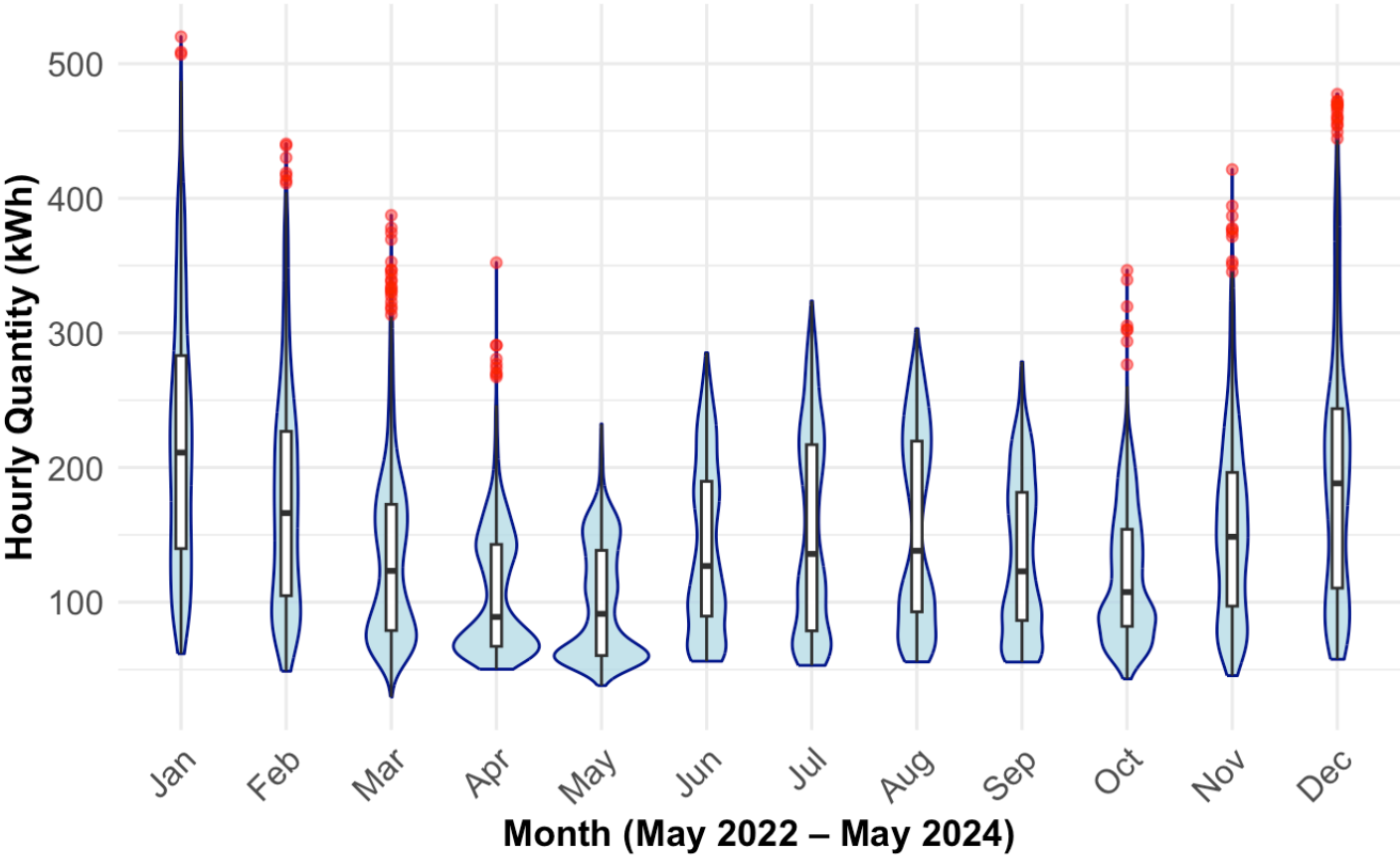
account_b_data$Datetime_parsed <- as.POSIXct(account_b_data$Datetime, format = "%m/%d/%y %H:%M")
account_b_data$Hour <- as.integer(format(account_b_data$Datetime_parsed, "%H"))

ggplot(account_b_data, aes(x = Month, y = Quantity)) +
  geom_violin(fill = "lightblue", color = "darkblue", alpha = 0.7) +
  geom_boxplot(width = 0.1, fill = "white", outlier.color = "red", outlier.alpha = 0.5) +
  theme_minimal() +
  scale_x_discrete(labels = month.abb) +
  scale_y_continuous(labels = scales::comma) +
  theme(
    axis.text.x = element_text(size = 12, angle = 45, hjust = 1),
    axis.text.y = element_text(size = 12),
    axis.title.x = element_text(size = 13, face = "bold"),
    axis.title.y = element_text(size = 13, face = "bold"),
    plot.title = element_text(size = 14, face = "bold"),
    plot.subtitle = element_text(size = 11, color = "gray40"),
    plot.caption = element_text(size = 10, color = "gray50")
  ) +
  labs(
    title = "Distribution of Hourly Energy Consumption by Month (Account B)",
    subtitle = "Violin shows full distribution; box shows median and IQR; red dots are outliers",
    x = "Month (May 2022 – May 2024)",
    y = "Hourly Quantity (kWh)",
    caption = "Each observation represents one hourly energy reading"
  )

```

Distribution of Hourly Energy Consumption by Month (Account B)

Violin shows full distribution; box shows median and IQR; red dots are outliers



Each observation represents one hourly energy reading

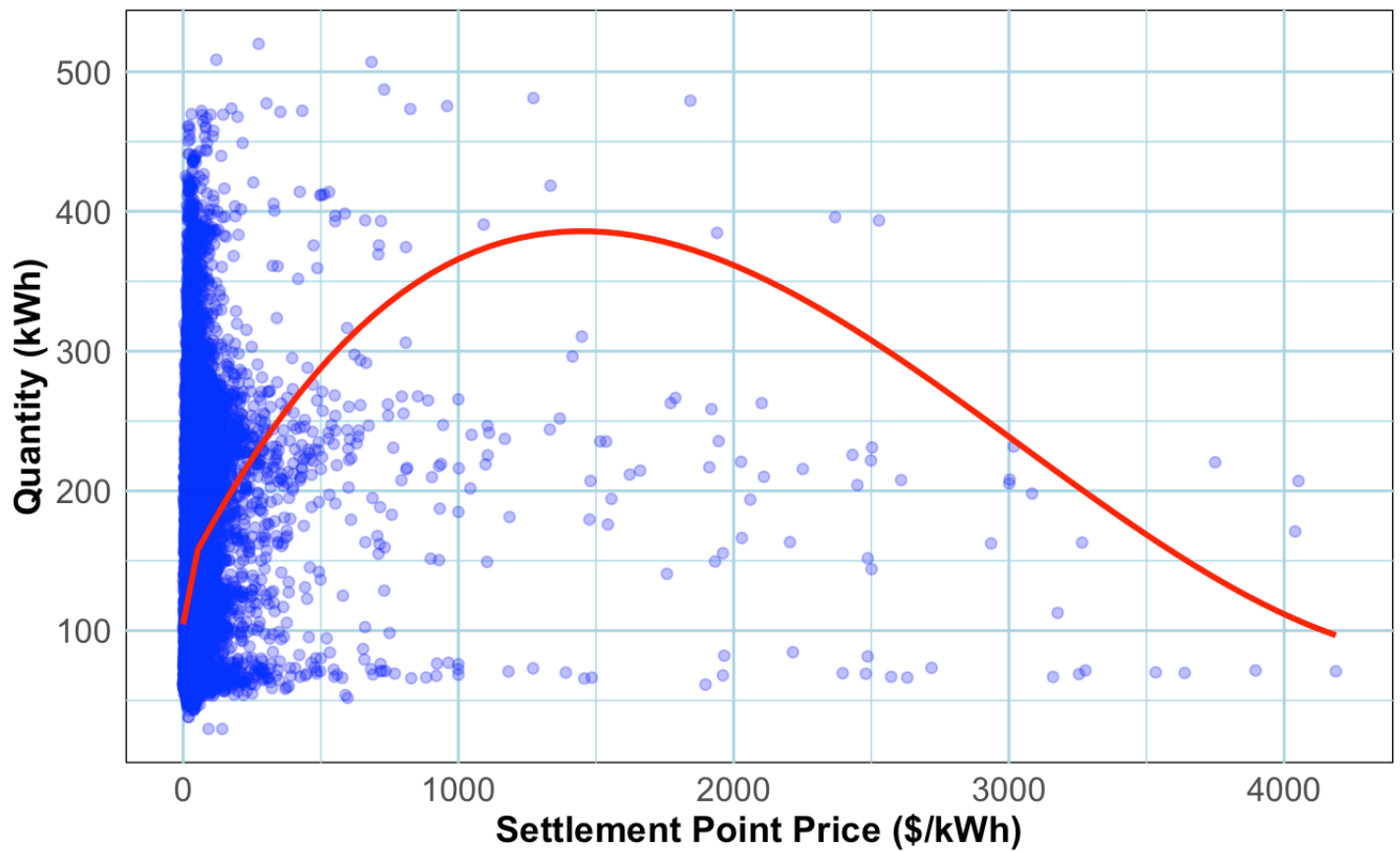
```

ggplot(account_b_data, aes(x = SettlementPointPrice, y = Quantity)) +
  geom_point(color = "blue", alpha = 0.3) +
  geom_smooth(method = "loess", color = "red", se = FALSE) +
  theme_minimal() +
  theme(
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white"),
    panel.grid.major = element_line(color = "lightblue"),
    panel.grid.minor = element_line(color = "lightblue"),
    axis.text.x = element_text(size = 12),
    axis.text.y = element_text(size = 12),
    axis.title.x = element_text(size = 13, face = "bold"),
    axis.title.y = element_text(size = 13, face = "bold"),
    plot.title = element_text(size = 14, face = "bold"),
    plot.subtitle = element_text(size = 11, color = "gray40"),
    plot.caption = element_text(size = 10, color = "gray50")
  ) +
  labs(
    title = "Energy Consumption vs. Settlement Price (Account B)",
    subtitle = "Red line shows LOESS trend; each point represents one hourly observation",
    x = "Settlement Point Price ($/kWh)",
    y = "Quantity (kWh)",
    caption = "Note: Price converted from $/MWh to $/kWh"
  )

```


Energy Consumption vs. Settlement Price (Account B)

Red line shows LOESS trend; each point represents one hourly observation



Note: Price converted from \$/MWh to \$/kWh